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Motion-Based Consequence Analysis: From Pre-Operation Risk Assessment to Live Operational Monitoring

By Frode Bloch

Brunvoll AS

Abstract

Current DP rules and guidance specify capability analysis in detail but treat consequence analysis (CQA) as little more than a periodic alarm. The prevailing requirement — recompute every five minutes, raise a flag only once a limit is breached — leaves the operator with no continuous indication of the probability of exceeding operational limits, in the intact condition or following a worst-case failure (WCF), and no view of the motion consequences of such a failure before it happens. The risk that CQA is meant to manage is inherently dynamic; the rules treat it as static.

This work develops a motion-based CQA framework that closes that gap end-to-end. A pre-operation risk assessment extends the conventional capability analysis with an estimate of vessel response and post-WCF excursion from the planned sea state, allowing the worst-case failure design intent (WCFDI) to be evaluated in motion terms. The same model is reformulated for online execution, producing a continuously updated CQA panel that estimates intact and post-WCF station-keeping footprint directly from live measurements and the DP model, without a runtime sea-state input. The panel is further extended to motion-sensitive mission equipment such as walk-to-work gangways.

Validation against a nonlinear DP simulator is complete; validation against logged operations is in progress.

Abbreviations and Definitions

CQA Consequence Analysis: the DP-rules-mandated periodic computation that flags whether the vessel can maintain position after a worst-case failure. Referred to throughout as the **classical CQA** where it needs to be distinguished from the extension proposed here.

Motion-based CQA The extension of the classical CQA proposed in this paper, which quantifies the **probability of exceeding operational limits**, in both intact and post-WCF conditions, continuously and in motion terms.

P50 / P90 / P95 Percentiles of the predicted maximum motion excursion. P_{50} is the typical value, exceeded in about half of operating windows; P_{90} is exceeded in roughly one window in ten; P_{95} is a conservative value exceeded in one window in twenty.

WCF Worst-Case Failure: the failure event (typically loss of a thruster group or bus section) against which CQA is evaluated.

WCFDI Worst-Case Failure Design Intent: the specific failure configuration the capability and consequence analyses are evaluated against; the vessel is designed to survive this.

DP model The mathematical model of the vessel used internally in the DP control system to estimate the vessel response from the combination of thruster action, sensor measurements, and position reference measurements. It carries estimates of position, velocity, and the slowly varying bias that absorbs the environmental load not otherwise modelled.

Capability analysis The classical pre-operation deliverable computing the achievable envelope of environmental force the vessel can hold against.

WCFDI decision matrix A traffic-light forecast of intact and post-WCFDI operability as a function of (vessel heading, forecast time), delivered before the operation starts.

Introduction

Consequence analysis (CQA) has been a requirement for redundant DP classes for several decades. The IMO consolidated the requirement at international level in MSC/Circ.645 in 1994 [1], and it is carried into the class rules [2, 3] as a mandatory software function on DP-2 and DP-3 vessels. Its purpose is straightforward: given the worst-case failure (WCF) the vessel is designed against, verify continuously that the DP system as a whole — sensors, thrusters, power, and control — can withstand the average prevailing environmental forces after the failure. If it cannot, an alarm is raised.

That obligation is met today by a periodic force-and-moment reserve check with alarm-on-breach. The check is defensible: applied with a percentage capacity margin it keeps the operator comfortably inside a safe envelope for the majority of operations. What it does not report is **motion**. The CQA does not indicate how the vessel will move in the intact condition, how it will move during and after a WCF, or how mission-sensitive equipment such as walk-to-work gangways, heave-compensated cranes, and ROV LARS will move with it. It does not quantify the probability of exceeding operational limits before those limits are breached. Modern DP operations increasingly run at sea states where the motion of the vessel or its mission equipment sits close to operational limits well inside the capacity-safe envelope, and the operator has no continuous, graded indication of proximity to those limits.

This paper proposes a **motion-based consequence quality analysis** (CQA) framework that closes that gap by adding a layer on top of the current CQA implementation, in two coupled deliverables: a **pre-operation risk assessment**, evaluating the worst-case failure design intent (WCFDI) in motion terms from the forecast sea state; and a **live CQA panel**, producing continuously updated P_{50} / P_{95} motion-footprint envelopes for intact and post-WCF conditions from measurements every DP-2/DP-3 vessel already publishes. The framework is additive: the current CQA continues to run and continues to raise its Drift Off / Force Off alarms; the motion layer adds one new operator alert, **Large Transient**, that fires when the motion prediction itself should be read as a lower bound rather than an upper bound. With the motion layer in place the operator can work closer to the operational limit in a controlled way, seeing proximity to the limit rather than waiting for a breach.

The remainder of the paper is organised as follows. Section 2 states the design goals of the framework. Section 3 develops the pre-operation motion capability analysis. Section 4 develops the live CQA panel and the two-channel design. Section 5 presents simulator-based validation on a CSOV benchmark vessel. Section 6 reports the in-progress validation against real operational logs. Section 7 concludes.

Goal

The motion-based CQA framework is designed to two operational goals. Both are stated from the perspective of the operator planning and running an offshore operation — for example a walk-to-work personnel transfer or a heave-compensated construction lift — where the station-keeping capacity is comfortable but the **motion** of vessel and mission equipment may sit close to the operational limits.

1. **Verify motion limits before the operation.** Before committing to the operation, the operator can evaluate whether the planned vessel and mission-equipment motion — both in the intact condition and following the worst-case failure (WCF) — will stay inside the operational limits at the planned sea state, heading, and operating window. The classical capability analysis already verifies that the DP system has the capacity to hold; the motion-based pre-operation risk assessment answers the next question, whether the resulting motion is inside the limits that actually constrain the operation.
2. **Maintain continuous awareness of the safety margin towards those motion limits during the**

operation. Once the operation is in progress, the operator continuously sees how close the vessel and mission-equipment motion sits to the operational limits, both in the intact condition and in the post-WCF condition. Rather than the binary alarm-on-breach of the classical CQA, the operator reads a continuously updated margin: comfortable, closer, at the limit. The awareness covers the post-WCF condition explicitly, so the operator sees whether the operation is inside the survivable motion envelope, not only inside the capacity envelope.

Four design principles support these two goals and are worth stating briefly up front so the methodology sections that follow can be read against them. First, **motion-based throughout**: pre-operation and live outputs are stated in vessel and mission-equipment motion, in the same units and thresholds the operator already reads on the DP console. Second, **one model in two regimes**: the same underlying motion model serves the pre-operation stage (fed by the forecast sea state) and the live stage (fed by live measurements and the DP model), so that the operator sees the same numbers before and during the operation. Third, **readily available measurements only, online**: the live layer consumes only signals every DP-2/DP-3 vessel already publishes, plus vessel-constant parameters — no runtime seakeeping model, no runtime spectral partition, no runtime wave forecast — so that the design stays defensible under mixed wind-sea and swell, cross-current forcing, and near-installation shielding. Fourth, **additive to the classical framework**: the motion layer sits on top of the classical capability + CQA computation and consumes its outputs, without replacing or contesting any rule-mandated behaviour; the classical Drift Off and Force Off alarms remain the primary hard-failure signals.

Motion-Based Capability Analysis

Why motion matters in modern DP operations

Consider a walk-to-work operation transferring personnel to a wind turbine. Once the gangway has landed, its tip is not free to follow the vessel: it rests on, or is pressed against, a landing point on the structure, and is therefore held fixed in the structure's frame. The vessel motion underneath is absorbed instead by the gangway mechanism — principally telescope stroke, together with slew. Both have hard mechanical end-stops. A gangway carries only a limited amount of stroke between fully extended and fully compressed, and that stroke budget, shared with the slew envelope, is what sets the excursion the gangway pedestal — and hence the vessel — can be allowed before the mechanism runs out of travel and the transfer must be stopped. IMCA M254 [4] sets out how station-keeping footprint limits should be established for an operation; the resulting warning and alarm radii are what express this allowance in a form the DP operator can work to. Throughout this paper we use 2 m and 4 m as representative warning and alarm radii for the benchmark vessel. They are illustrative values for that vessel and gangway, not fixed limits: the radii are vessel-, equipment- and operation-specific and are established per operation.

The consequence is that the binding limit is a **motion** limit that sits well inside the capacity-safe envelope. In many sea states the DP capacity margin against wind, wave, and current is generous and the classical CQA reports green, yet the transfer stops because the vessel's motion footprint has consumed the gangway's mechanical travel. The same picture recurs in heave-compensated construction, ROV LARS operations, and offshore crane work: the operational limit is a motion limit on mission-sensitive equipment, not a capacity limit. The IMCA M254 bands are themselves a motion concept: the yellow and red station-keeping footprint limits are distances the vessel must stay within, not percentages of capacity reserve. What a motion-based pre-operation risk assessment adds is the **probability of exceeding** them — quantiles of the motion process evaluated against those limits — for the vessel and, through the same shared kinematic model, for the mission-equipment quantities the operator actually cares about, such as telescope stroke.

High-quality weather forecasts as an enabler

The motion-based pre-operation risk assessment is only defensible because the meteorological input is now good enough. Modern global ensemble products such as the ECMWF Integrated Forecasting System [5] deliver directional wave spectra with useful skill at 24 to 72 hours horizon for the wind-sea component, together with a probabilistic view from the ensemble spread. Regional hindcast reanalyses such as NORA3 [6] provide a directional spectrum climatology at 3 km resolution for the North Sea and Norwegian Sea that has been extensively validated against buoy measurements and is a useful cross-check against archived forecasts. Skill on the swell component and on rapidly evolving weather systems remains the harder part of the problem, and cross-basin swell propagation and mixed multi-modal sea states are known weak points; but the forecast directional spectrum is credible enough that using it as the input to a motion-response model produces an operationally meaningful envelope rather than a numerical artefact. The pre-op layer of the framework treats the forecast directional spectrum as its primary input and propagates it through the vessel response model described in the next subsection.

Method: from forecast sea state to motion envelope

The pre-operation method takes the forecast sea state as input and produces a motion-footprint envelope — a probability distribution of peak excursion over the operating window — for both the vessel and any user-defined point on mission-sensitive equipment. It comprises five steps.

a. Vessel-response model. A 3-DOF linearised model of the DP station-keeping regime,

$$M\ddot{\eta} + D\dot{\eta} = \tau_{env} + \tau_c, \tag{1}$$

with $\eta = (x, y, \psi)^\top$ the position and heading deviation from setpoint, M the mass and added-mass matrix, D the linear damping, τ_c the DP thrust command from the vessel's own PID + integrator controller closed around η, and τ_{env} the environmental forcing. Only vessel mass, damping, and controller tuning are required; a full 6-DOF seakeeping representation is not used because the pre-op deliverable is a horizontal-plane station-keeping footprint.

b. Environmental forcing. From the forecast directional wave spectrum, mean wind, and mean current, τ_{env} is composed of three components: the second-order wave-drift force per DOF, obtained by applying the vessel's quadratic transfer functions (QTFs) [7] to the directional spectrum, and the wind and current forces, computed with the same models the vessel's DP control system uses. The wave-drift term dominates the LF driving spectrum; wind and current appear primarily as bias offsets absorbed by the DP model.

c. From τ_{env} to motion envelope. Under Eq. (1), the intact response has a low-frequency component driven by wave-drift and wind and a wave-frequency component driven by first-order wave excitation, both approximately stationary over the operating window.

What the operator needs from this is the answer to a simple question: how far is the vessel likely to move, at worst, during the operation? The quantity of interest is therefore the **maximum position excursion over a selected operating window** R — the largest distance the vessel wanders from its setpoint during, typically, the next 30 minutes. That maximum is not a fixed number: it varies from one realisation of the sea state to the next, so the framework treats it as a random quantity. Characterising it from the response statistics is standard extreme-value practice for stationary processes; see Rice [8] and Vanmarcke [9] for the derivations.

What is then reported is not the distribution itself but the number the operator can act on: the probability that the excursion exceeds the warning and alarm limits set for the operation. Alongside it, two representative

excursions give a sense of scale — a typical value, exceeded in about half of operating windows, and a conservative value, exceeded in only one window in twenty.

The traffic-light band that decides operable / caution / no-go is set by comparing a high excursion level against the vessel- and mission-specific warning and alarm radii established per the IMCA M254 station-keeping alarm concept [4]. The level used is the one exceeded in about one operating window in ten, written P_{90} in what follows and in the figures; on the same convention P_{50} is the typical excursion, exceeded in about half of operating windows, and P_{95} a conservative one exceeded in one window in twenty. The vessel's wave-frequency motion is combined with the slower drifting motion to give the total excursion.

d. Extension to mission equipment. The same excursion process, projected through mission-equipment lever arms and coupling kinematics, gives the motion at any point of interest — a LARS boom head or crane hook — as a linear combination of vessel translations and rotations. For a landed gangway the quantity of interest is not a point motion but the travel the mechanism must absorb to hold the tip fixed on the structure: the same projection yields the telescope stroke deviation directly, which is what the mechanical end-stops constrain. No new random inputs are introduced; vessel and mission-equipment footprints appear together in the pre-op operator summary.

e. Note on conservatism. The extreme-value result is mildly conservative for a feedback-controlled vessel, because the controller suppresses the largest excursions the underlying theory allows for. The predicted excursion therefore comes out somewhat larger than the vessel actually experiences; the simulation studies quantify the margin.

Assumptions

The pre-operation motion model rests on the following explicit assumptions.

- The sea state is stationary over the operating window, so that process statistics (variance, spectral moments) are constant.
- The forecast directional spectrum is representative of the actual sea state during the window, within the skill of the forecast product used.
- The vessel QTFs, wind coefficients, and current coefficients are the design-intent values for the vessel and heading; drift and loading variation are absorbed into the framework's conservatism margin.
- Forward-speed effects are absent (station-keeping regime): the vessel is dynamically holding a setpoint, not transiting.
- The low-frequency and wave-frequency responses are treated as independent and combined by root-sum-square.
- The classical capability analysis provides the outer envelope of operability. The motion-based envelope produced here is the inner envelope: within the capacity-safe outer envelope, the motion envelope identifies where operational limits — vessel or mission equipment — are first exceeded. The two envelopes are used together, not one instead of the other.

Example: intact operability polar

The framework produces two kinds of polar, answering different questions. The **design polar** of this subsection and the next sweeps wind speed with the sea state growing alongside it, and answers whether a given vessel and thruster layout is suitable for a work scope at all. It is a design and feasibility chart of the same family as the standard DP capability plot, not something a navigator reads on the bridge. The

forecast-conditioned polar of the following subsection pins the sea state to a specific forecast and varies only the wind, and is the operational chart. Both are built from the same motion model.

Figure 1 illustrates the intact-condition operability polar produced by the framework for the CSOV benchmark vessel introduced in Section 5. At any given vessel heading and mean wind speed the polar answers whether the operation is operable (green), in a caution band (yellow), or no-go (red) in terms of the intact motion footprint. The colour band is driven by the P90 of the running-maximum excursion against a warning and alarm radius: for the vessel base position the warn / alarm thresholds are the station-keeping radii established for the operation (here 2.0 m / 4.0 m); for the gangway telescope they are derived from the mechanism's remaining travel. The benchmark gangway telescopes between 18 m and 32 m and is landed at mid-stroke, leaving 7 m of travel in each direction; warning and alarm are set at 60% and 80% of that remaining travel, giving 4.20 m and 5.60 m. Landing the gangway off-centre reduces the travel available in one direction and tightens these thresholds accordingly. The sea state at each polar radius is generated from the wind speed using the DNV-ST-0111 [10] wind-wave correlation for a fully-developed sea, which is the Pierson-Moskowitz relation, with a Bretschneider spectrum shape and a fixed current component. The operating window is 20 minutes.

Two features are worth calling out. First, the operable envelopes are visibly anisotropic: both panels show larger operable radii in head and following seas than in the beam sector, as expected from the projection of environmental force through the vessel geometry. Second, the gangway telescope panel has a larger operable envelope than the vessel base position, because the gangway threshold levels are set from the mechanical stroke of the telescope, not from a station-keeping alarm limit; the vessel and mission-equipment footprints are treated in the same motion-based framework but each against its own operational limits. The framework can produce the same polar for any additional mission-equipment point that the operator provides thresholds for.

Example: post-WCFDI operability polar

Figure 2 overlays the post-WCFDI no-go boundary (black dashed) on the same intact-prior polar of the previous subsection. The WCFDI scenario for this illustrative figure is a loss of one of three symmetric thruster groups, which for the standalone prototype used to generate the figure is parameterised as a per-DOF surviving-fleet capacity of two-thirds of intact and an allocator reallocation time $T_{\text{realloc}} = 10$ s; the P90 Gaussian factor is $k_\sigma = 0.674$. In a real deployment the per-DOF residual capacity is not a scalar of intact — the motion layer consumes whichever $\tau_{\text{residual, cap}}$ the classical CQA already publishes over the survivor set. The dashed boundary is the wind-speed contour at which the P90 of the running-maximum excursion, evaluated on the post-WCFDI closed-loop dynamics with this residual capacity, reaches the alarm radius per DOF.

The important operational message is that the post-WCFDI boundary sits well inside the intact alarm ring — the vessel panel shows the WCFDI no-go contour at roughly 15-20 m/s wind speed in the beam sector, compared to the intact alarm at around 25 m/s. This is the motion equivalent of the tighter capacity envelope the classical CQA already screens against, but expressed in the operator's motion units rather than in a capacity-reserve percentage. Two features of the plot deserve comment. First, the WCFDI restriction is materially asymmetric across headings: the boundary is tightest in the beam sector (heading 90 / 270°) where the DOF that lost the most capacity is loaded worst, and closest to the intact envelope in head and following seas. Second, the WCFDI restriction bites less hard on the gangway telescope panel than on the vessel base position panel: the mission-equipment threshold is a mechanical stroke limit that is generous relative to the post-WCFDI excursion at the same sea state. For this vessel and gangway configuration the vessel-station-keeping metric is the binding constraint after the WCFDI, not the gangway telescope.

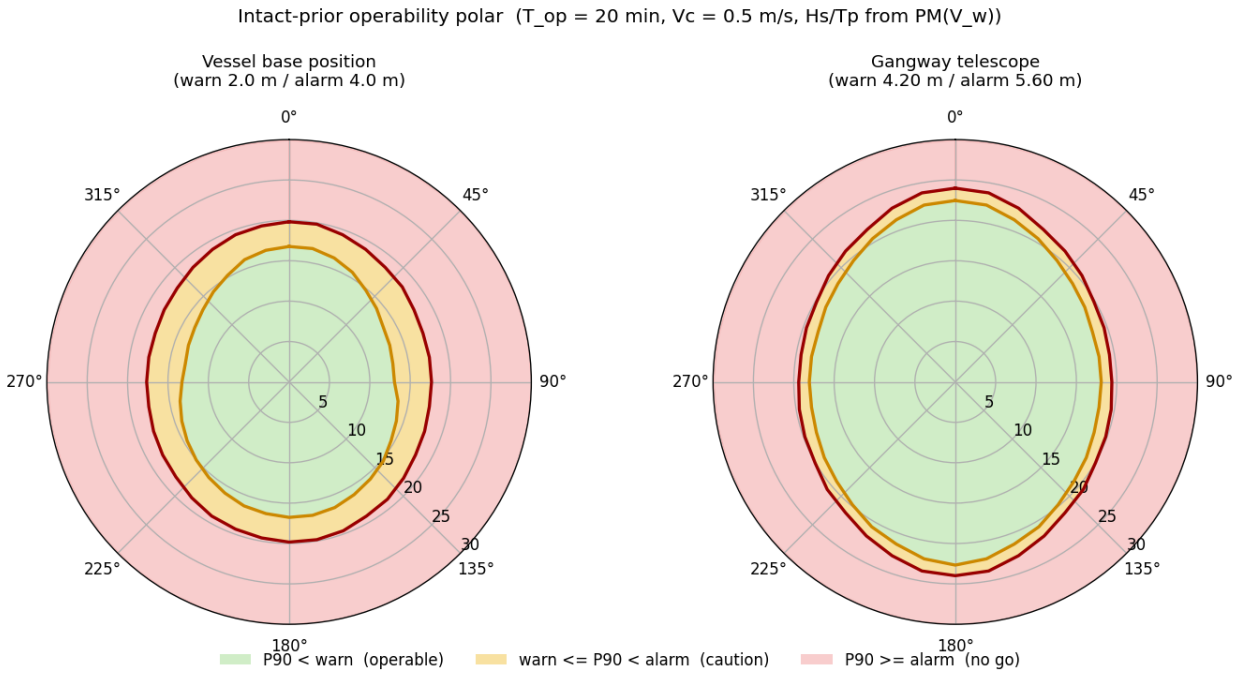


Figure 1: Intact-condition operability polar for the CSOV benchmark vessel and forward walk-to-work gangway. Polar radius is mean wind speed [m/s]; heading is vessel-relative wave direction. Colour band is driven by the P90 of the running-maximum motion excursion against the vessel / mission-equipment warning and alarm radii.

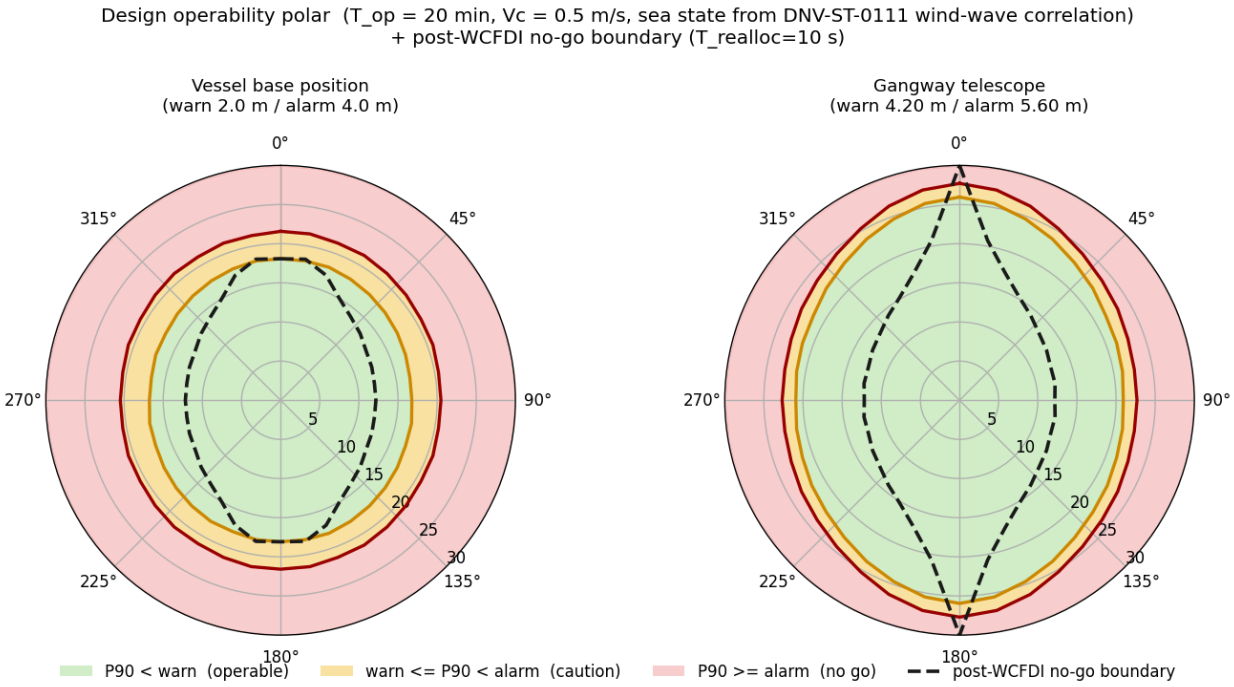


Figure 2: Post-WCF operability polar for the CSOV benchmark vessel and forward walk-to-work gangway. The coloured bands are the intact-condition operability of Figure 1; the black dashed contour is the wind speed at which the alarm radius would be exceeded about one operating window in ten after the worst-case failure.

Example: forecast-conditioned operability polar

The design polars answer a design question. The operational question is different: given the sea state actually forecast for the operating window, which vessel headings work, and how much margin is left on each? Figure 3 answers it directly. The wave condition and the current are pinned at their forecast values and only the wind speed is swept, so the boundary for each vessel heading is the wind speed at which the motion limit would be reached in **this** sea state. The forecast wind speed is drawn as a ring on the same axes, and the radial gap between that ring and the boundary is the wind margin the operator has in hand at each heading.

The example is a Beaufort 6 forecast — the nominal DNV-ST-0111 condition, Hs = 3.1 m, Tp = 8.5 s, Vc = 0.75 m/s — with the wind veered 90° ahead of a sea that has not yet turned to follow it, and the tidal stream running with the wind. This is an ordinary forecast situation rather than a contrived one, and it is what makes the envelope genuinely asymmetric: the heading that is most comfortable for the waves is not the one that is most comfortable for the wind and current.

The operational content is the margin, not the colour. On the vessel panel the caution boundary ranges from 19.3 m/s to 24.6 m/s across headings, so against a forecast wind of 13.8 m/s the operator has between 5.5 and 10.8 m/s of wind in hand depending on how the vessel is oriented — roughly a factor of two between the tightest heading and the most open one. An operator reading only a pass / fail capability check sees "operable" across the whole circle and has no way to tell that some headings carry twice the margin of others. That margin is what makes the difference between an operation that absorbs a forecast busting by a few metres per second and one that does not.

Each panel carries two further contours, and the distinction between them is the same one the live panel draws. The black dashed curve is the post-WCF alarm boundary: the wind speed at which the motion after the worst-case failure would cross the alarm limit. The dotted curve marks where a **large transient** becomes possible after the failure: beyond it the post-WCF motion figure should be read as a lower bound rather than a tight prediction. It is the pre-operation counterpart of the Large Transient alert on the live panel, reached by a different computation — a bistability score on the predicted trajectory here, the saturation probability p_{sat} there — but carrying the same operational meaning.

The two say different things here, and that is the point. On motion alone the operation is comfortable: the post-WCF alarm boundary does not fall below 16.0 m/s at any heading, against a forecast wind of 13.8 m/s, so no heading is predicted to breach the envelope. But the large-transient boundary falls to 11.6 m/s in the worst orientations, and at 10 of the 36 headings evaluated it sits below the forecast wind. In those orientations — two sectors around 020-060° and 200-240° — the framework is not saying the vessel will exceed its motion limit after a failure; it is warning that a large transient is possible and that the predicted motion may therefore be an underestimate. An operator reading a single number would see a comfortable margin, with no indication that a quarter of the heading circle carries that risk.

The two panels also show which limit is actually binding. The gangway telescope boundary does not fall below 31 m/s at any heading, and at some orientations the telescope limit is not reached at any wind speed of interest because the vessel motion barely projects onto the telescope axis. For this vessel, gangway configuration and forecast the vessel station-keeping limit binds everywhere and the gangway has substantial margin behind it — but that is a result of the geometry rather than a general rule, and a shorter-stroke gangway or an off-centre landing would move the binding limit. Both limits are motion limits; the classical force-balance check is a separate constraint and appears as the top panel of the decision matrix in the next subsection.

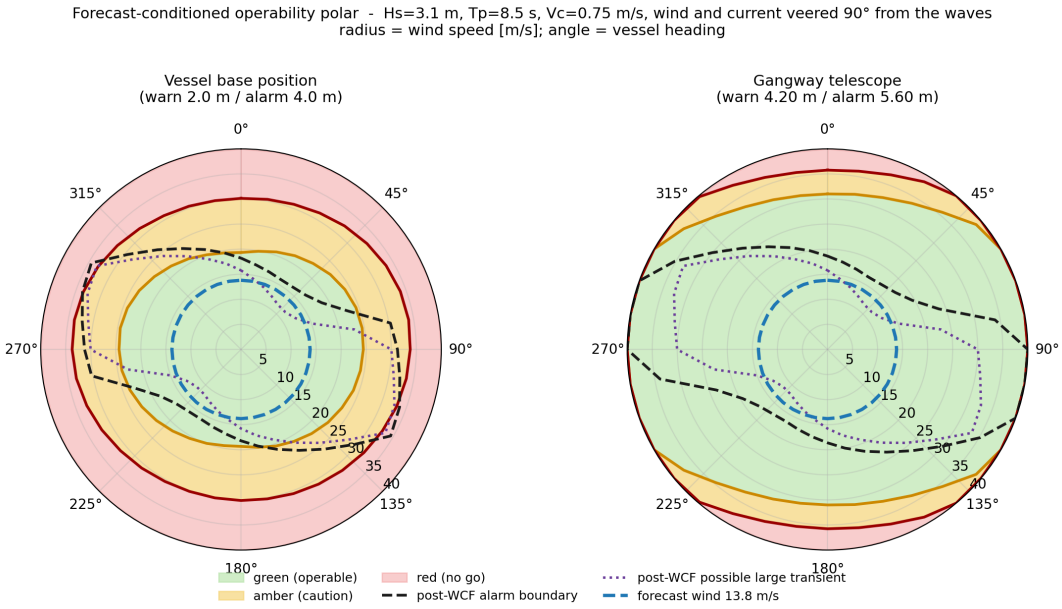


Figure 3: Forecast-conditioned operability polar for the CSOV benchmark, vessel base position (left) and gangway telescope (right) on a common scale. The wave condition and current are pinned at the forecast values (Hs = 3.1 m, Tp = 8.5 s, Vc = 0.75 m/s, wind and current veered 90° from the waves) and only the wind speed is swept. Radius is wind speed; the angular coordinate is vessel heading. Green is operable, the amber band runs from the warning to the alarm boundary, and the dashed ring is the forecast wind speed. The radial gap between the ring and the green boundary is the wind margin available at that heading. The black dashed contour is the post-WCF alarm boundary; the dotted contour marks where a large transient becomes possible after the failure, beyond which the post-WCF motion figure should be read as a lower bound rather than a tight prediction. Gangway boundaries beyond the plotted range are clipped at the rim: at those headings the telescope limit is not reached at any wind speed of interest.

Example: WCFDI decision matrix

The operability polar of the previous subsection gives a static picture: for a given sea state, which headings are operable? The WCFDI decision matrix (Figure 4) puts that picture on a forecast timeline. Each cell of the matrix is one (heading, forecast slot) combination; the cell colour is the utilisation ratio evaluated on the forecast sea state at that time, using the same green→yellow→red gradient across all three panels so that different views of operability can be read directly against each other. The example is driven by a synthetic 24-hour storm.

The three panels correspond to three different consequence-analysis questions asked at the same (time, heading) cell:

- **Classical CQA utilisation** (top panel): the ratio of mean environmental force to residual (post-WCFDI) thrust capability, per DOF, taken maximum across DOFs. This is what the current force-and-moment reserve check that runs on any DP-2/DP-3 vessel reports. The operational bands used by the reference DP system are caution at 80% utilisation and warning at 90%.
- **Motion-based intact utilisation** (middle panel): the ratio of the intact-condition P90 running-maximum footprint to the operation's alarm radius. Reaches 1.0 when the intact motion of the vessel itself would cross that alarm threshold.
- **Motion-based post-WCFDI utilisation** (bottom panel): the same ratio computed on the post-WCFDI motion envelope, i.e. what the motion would be if the worst-case failure occurred at that (time, heading) cell.

The operator's task is to plan a safe operation through the forecast window. The three panels answer that task in three different ways. The classical CQA panel identifies the operational constraint the current rules already enforce: a narrow yellow-orange caution band and a small red spot at the storm peak (12:00) in the two beam sectors, driven by the environmental force approaching the residual capability. An operator running only the classical CQA would read this as "the storm peak is borderline in beam headings; work around it and the operation is fine". The motion-based intact panel adds essentially nothing new here — the vessel motion itself never comes close to the alarm radius, so the classical CQA plus intact motion confirms the same operational verdict.

The motion-based post-WCFDI panel changes the picture. Even though the classical CQA reports at most yellow-caution across the storm peak, the motion-based post-WCFDI utilisation reaches saturation (dark red, well past the alarm radius) in a narrow ~2-hour window around 12:00 in the beam headings. Physically, this is the regime where the storm energy is still comfortably below the vessel's survivable capacity but the **motion** consequence of a WCF at that instant would take the vessel outside the operational motion envelope. Two heading windows around head and following seas (roughly 150-210° and 330-030°) stay green even at the storm peak.

The operator's plan therefore depends on which panel is authoritative. On classical CQA alone the plan is "operate through the storm, easing off around 12:00"; on the motion-based post-WCFDI panel the plan becomes "avoid the beam sector 11:00-13:00 window entirely, or use one of the two green heading windows through it". The 2-hour narrow red band is the added operational content the motion-based framework delivers: the **when** and the **which heading** at the resolution the motion actually constrains, not just the **whether** at the resolution the residual capacity constrains.

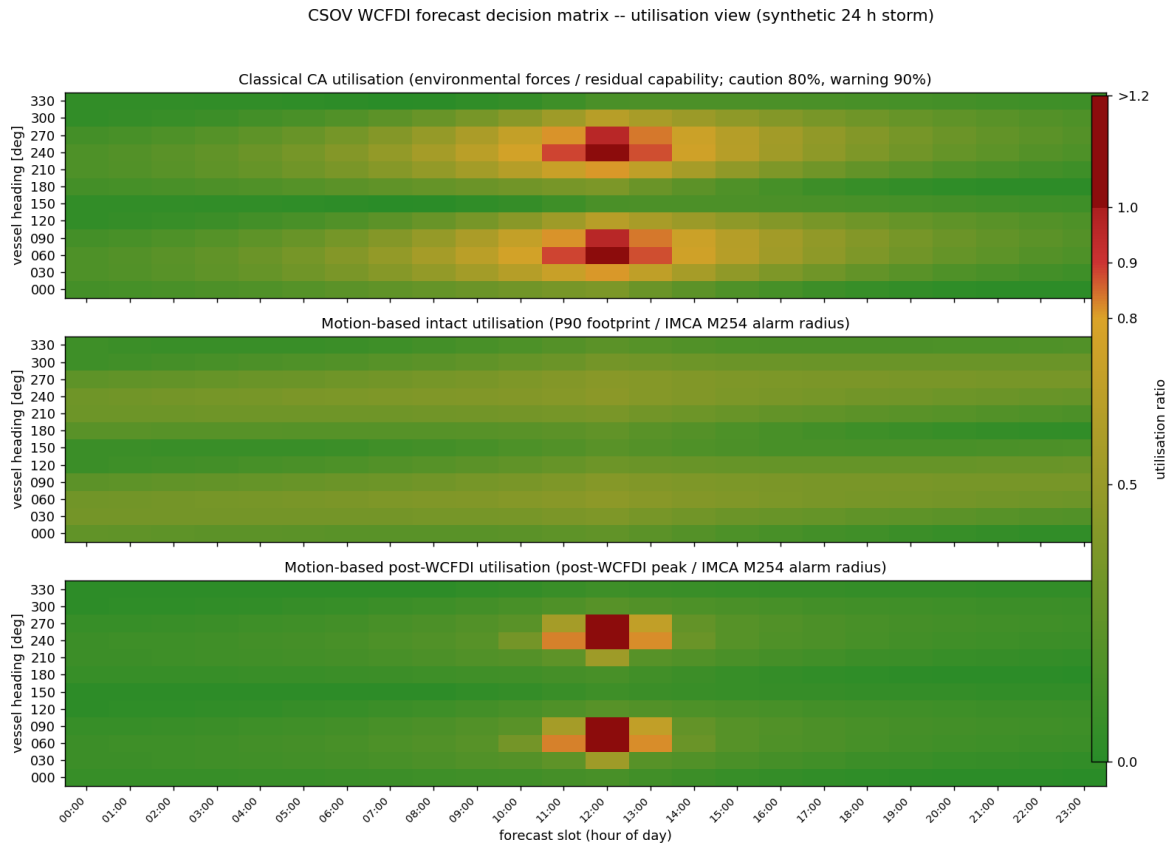


Figure 4: WCFDI forecast decision matrix over a synthetic 24-hour storm for the CSOV benchmark vessel, shown as three side-by-side utilisation heatmaps. Rows: vessel heading (deg, true). Columns: forecast slot (hour of day). All three panels use the same green→yellow→red gradient colour map on a common utilisation ratio scale 0 to 1.2 (colour bar right). Reference thresholds: for the classical CQA panel, caution at 0.80 and warning at 0.90 (reference DP system convention); for the two motion panels, caution at 0.50 (the warn / alarm radius ratio) and warning at 1.00 (P90 equals the alarm radius). Top: classical CQA utilisation (environmental forces / residual capability, max across DOFs). Middle: motion-based intact utilisation (P90 footprint / alarm radius). Bottom: motion-based post-WCFDI utilisation (post-WCFDI peak / alarm radius).

Example: pre-op operator summary

The polar plots and the decision matrix are planning views over sea state and time. The pre-operation operator summary (Figure 5) collapses a single operating point — the state selected for the operation — into a single-page display that the operator reads before committing to the operation. Two bars stack vertically: the top bar is vessel base-position capability, the bottom bar is gangway telescope capability, both against their own warn / alarm bands. Each bar shows the median (open circle, P_{50}) and the tail (filled diamond, P_{95}) of the post-WCF running-maximum motion excursion evaluated on the current operating state, with the traffic-light background driven by the same P90-based threshold used in the polars.

The important content on the display is not the traffic light itself — the operator already reads that on the polar — but the two headline probability numbers on each bar: "P(footprint > amber)" and "P(footprint > red)". In the example shown, the vessel bar reports "if WCF now: P(footprint > 2 m) = 5.0 %, P(footprint > 4 m) = 0.0 %". This is the concrete delivery of the abstract's **probability of exceeding operational limits**: not "the vessel can hold" (yes), not "the footprint is 2 m" (a single point estimate), but a probabilistic statement that the operator can weigh against the operational acceptance criterion for the specific mission. The gangway bar reports the analogous "P(end-stop hit)". Both are direct products of the same motion-based framework used to build the polars and the decision matrix; the operator sees one coherent set of numbers before and during the operation.

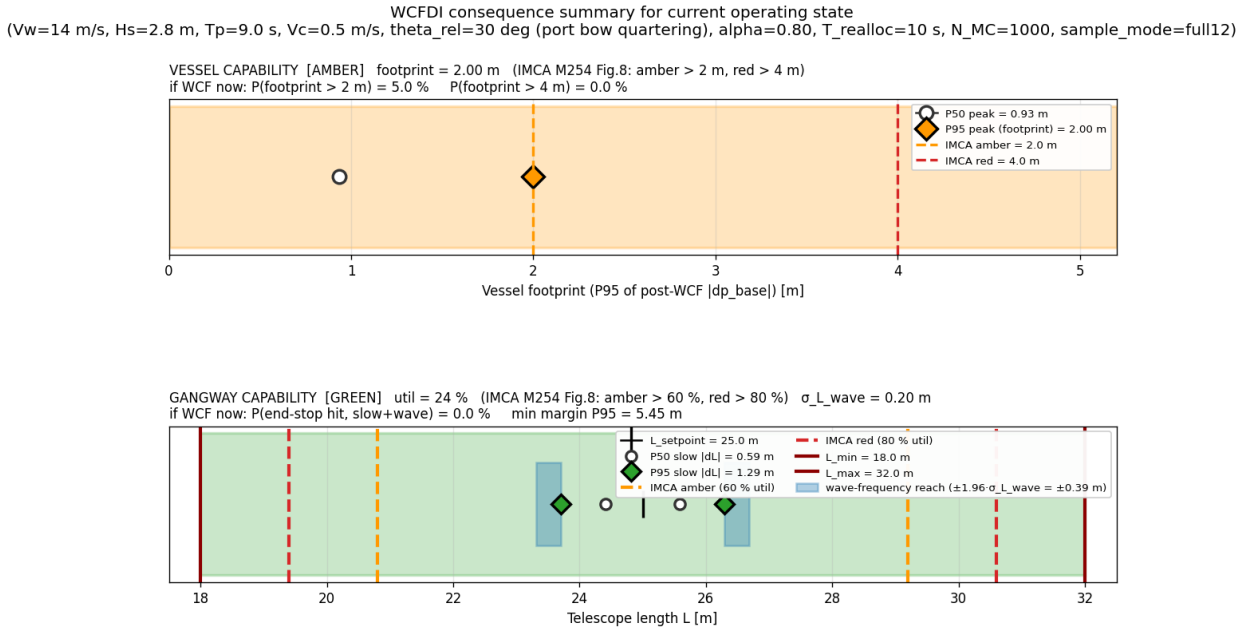


Figure 5: Pre-operation operator summary for the CSOV benchmark at a single operating point (Vw=14 m/s, Hs=2.8 m, Tp=9.0 s, bow-quartering seas, WCFDI scenario per Fig. 2). Top bar: vessel base-position capability. Bottom bar: gangway telescope capability. Open circle: P_{50}. Filled diamond: P_{95}. Vertical dashed lines: warn / alarm thresholds per system. Headline probabilities quantify the risk of exceeding each threshold under the post-WCF motion excursion.

Consequence Analysis (Live)

Positioning: additive to the classical CQA

The live CQA panel runs alongside — not in place of — the classical CQA required by IMO MSC/Circ.645 [1]. The classical CQA continues to compute and publish four quantities that the motion layer subscribes to: the per-WCF-candidate capacity reserve, a drift-off flag per DOF (the survivor set has zero capability against the prevailing environmental force in that DOF), a force-off flag per DOF (the survivor set is available but the prevailing demand exceeds its cap), and τ_{lost}, the vector of thrust that would be lost in the WCF at each candidate failure. The motion layer treats these as its inputs, along with the DP model state (position and velocity deviation $\eta, \dot{\eta}$; bias estimate $\hat{b}$) and a rolling buffer of the allocator feedback τ_{fb}; it does not duplicate the capacity-reserve computation and it does not consume any sea-state input at runtime.

τ_{lost} is not an instantaneous loss that stays at zero: the survivor set is reallocated back toward the pre-failure demand as fast as the thruster inertia and the allocator update rate allow. Figure 6 sketches the profile the motion layer assumes. At the WCF event, the available thrust capability magnitude drops instantaneously by τ_{lost}, of which an immediately recovered fraction γ_{imm} is picked up straight away by the survivor set. From that step the survivor allocator then recovers the remaining shortfall exponentially with characteristic time T_{realloc}, asymptoting to one of two targets. In the **healthy** case, when the pre-failure demand τ_{pre} is inside the residual capacity cap $\tau_{\text{residual, cap}}$ — the per-DOF thrust envelope of the surviving fleet, published by the classical CQA — the curve recovers back to τ_{pre} and the classical CQA reports green. In the **saturated** case, when τ_{pre} exceeds $\tau_{\text{residual, cap}}$, the same exponential recovery saturates at the cap instead of at τ_{pre}; the classical CQA raises Force Off on that DOF, and the motion layer's saturation-probability channel raises the **Large Transient** alert on the transient itself. The motion layer consumes τ_{lost} and $\tau_{\text{residual, cap}}$ as-is from whatever classical CQA is running in the vessel — no vessel-specific capacity scalar is introduced. The one modelling parameter internal to the motion layer is the reallocation characteristic time T_{realloc}; for the CSOV benchmark used throughout this paper we take $T_{\text{realloc}} = 10$ s, consistent with modern DP allocators. This time profile is what drives both the ensemble evaluator and the transient channel.

An important reading of Figure 6 is that it plots the **available** thrust capability per DOF, not the thrust the DP controller commands. The DP controller can and does command transient forces above τ_{pre} to arrest the motion picked up during the drop, provided the survivor set has the capability to deliver them; that commanded-thrust overshoot is captured inside the closed-loop vessel model of the method section (the controller term $\tau_{\text{ctrl}}(t)$ in Eq. 1), not in the recovery curve here. The first-order form of the recovery envelope is a compact approximation of the survivor-allocator dynamics: the immediately recovered fraction γ_{imm} and the characteristic time T_{realloc} are calibrated against reference DP simulator runs, and the approximation is confirmed to be operationally sufficient by the simulation validation.

The rule-mandated Drift Off and Force Off alarms remain the primary hard-failure signals in the operator UX; nothing in the motion layer suppresses or replaces them. The motion layer adds one new operator-visible alert, **Large Transient**, that fires when the live allocator feedback approaches the survivor's cap and the linear motion prediction should be read as a lower bound rather than as a tight upper bound. The alert is designed to sit between the classical green state and the classical Force Off warning: it draws attention to the **approach** to a limit rather than to its breach. The mechanism is developed in the saturation-probability channel.

Extreme-value method for exceedance envelopes

The live panel reports the same quantity as the pre-op layer — the distribution of the running maximum of $|\eta_{xy}(t)|$ over the coming 30 minutes — using the same extreme-value results [8], [9]. What differs is where

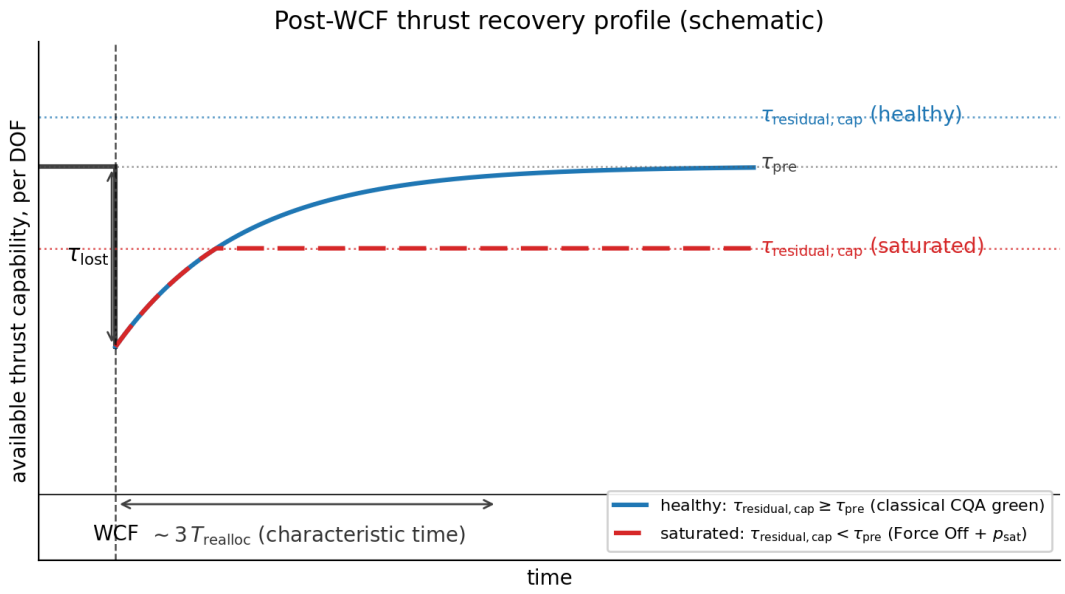


Figure 6: Post-WCF thrust recovery profile assumed by the motion layer, per DOF. At the WCF event the total commanded thrust capability drops instantaneously; the survivor set then recovers exponentially with characteristic time T_{realloc}, asymptoting either to the pre-failure demand τ_{pre} (healthy case, blue) or to the residual capacity cap $\tau_{\text{residual, cap}}$ when τ_{pre} exceeds the cap (saturated case, red-dashed). The saturated case is the operational territory of the classical Force Off alarm and of the motion layer's saturation-probability channel.

the inputs come from: instead of a forecast sea state, the statistics of the motion process are estimated online from the DP model's own position-deviation channel.

The panel maintains a recursive estimate of the position-deviation standard deviation on a sliding 10-minute window. The estimate is a distribution rather than a single value, so estimator uncertainty is visible to the operator as a widened envelope while the window fills. The estimator also accounts for the autocorrelation of the position samples, which would otherwise make the estimate look more confident than the data supports.

The panel reports P_{50} and P_{95} of the running maximum against the same vessel and mission-equipment warn / alarm radii used in the pre-op operator summary. The mild conservatism noted in the method section applies unchanged and is retained as margin. The intact-side output of the live panel is therefore directly comparable to the pre-op operator summary, but continuously updated on live data and with no sea-state input at runtime.

Motion after WCF: ensemble evaluator

The live panel needs a post-WCF motion prediction that is continuously updated on the vessel's actual state and can be trusted under the range of pre-failure conditions the WCF could catch it in. A single deterministic run of the vessel model from a point estimate of its steady state would collapse that range to a single number; the ensemble evaluator instead samples it. The idea is illustrated in Figure 7: rather than one trajectory, the panel propagates many, each starting from a recent snapshot of the actual vessel state, and reads its output from the spread across that ensemble.

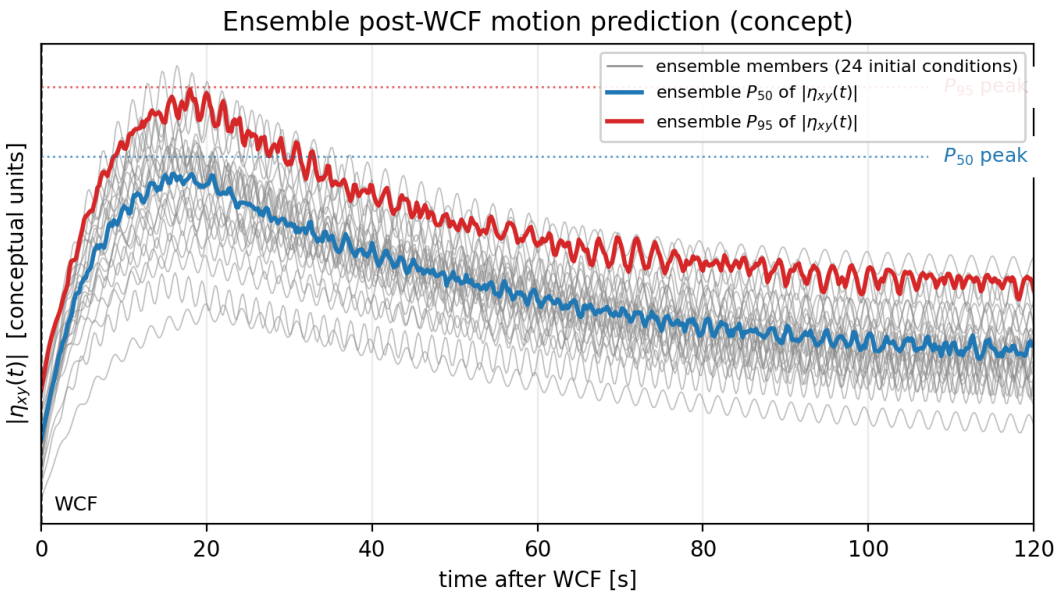


Figure 7: Conceptual illustration of the ensemble post-WCF motion prediction. At each panel update, snapshots of the recent vessel state are used as initial conditions for the vessel model, and each is propagated forward through the post-WCF window. The spread across the resulting trajectories is summarised by the ensemble P_{50} and P_{95} of $|\eta_{xy}(t)|$; the panel reports the P_{50} and P_{95} of the per-trajectory peaks (dotted lines). Numeric scales are omitted deliberately — the figure is illustrative of the mechanism, not of a specific vessel or sea state.

Concretely, the evaluator maintains a rolling 300-second buffer of vessel-state snapshots $\left(\eta, \dot{\eta}, \hat{b}, \tau_{\text{pre}}\right)$ drawn from the DP model and the allocator feedback. At each panel update, every snapshot in the buffer is used as the initial condition of the same closed-loop 3-DOF vessel model developed for the pre-op prediction in the method section (Eq. 1), propagated forward through the post-WCF window with τ_{lost} taken from the classical CQA output. Each propagation produces a peak-$|\eta_{xy}|$ trajectory; across the buffer this gives an empirical distribution of post-WCF peak excursions covering the actual pre-failure state variability the vessel is experiencing right now. The panel reports the median P_{50} and the tail P_{95} of that distribution, directly comparable to the pre-op operator summary.

Using the same vessel model the pre-op layer uses is deliberate: it enforces the "one model, two operating regimes" design goal of the Goal section. The two sides differ only in what feeds them — the pre-op side takes a forecast sea state through a vessel-response model to build the τ_{env} history; the live side takes the live $\hat{b}$ and τ_{pre} samples directly, without ever inferring a sea state. The classical CQA outputs τ_{lost} and the residual capacity feed both sides in the same way.

The ensemble evaluator handles the **long-term** post-WCF regime well because the closed-loop vessel model is quantitatively accurate for the non-saturated portion of the response. When the post-WCF residual allocator saturates, the linear model is no longer sufficient, and the motion prediction becomes a lower bound. That case is handled by the saturation-probability channel developed next.

Transient right after WCF: saturation probability channel

The linear closed-loop vessel model that drives the ensemble evaluator is quantitatively accurate as long as the post-WCF residual allocator does not saturate. Once the survivor set is forced onto its capacity cap, the linear model breaks down and the ensemble output becomes a lower bound on the true motion — no longer a tight prediction, and therefore no longer safe to display as the operator's motion number. The saturation-probability channel is an independent live gate that tells the operator when this happens.

The mechanism is direct. The panel buffers the live allocator feedback $\tau_{\text{fb}}(t)$ on a rolling 300-second window and low-pass-filters out the wave-frequency component to isolate the LF part that the residual allocator would have to absorb after a WCF. For each DOF a mean μ and standard deviation σ are computed, and a saturation probability

$$p_{\text{sat}} = \max_{\text{DOF}} \Pr\big(|\mathcal{N}(\mu, \sigma)| > \tau_{\text{residual, cap}}\big) \tag{2}$$

is evaluated against the per-DOF residual cap $\tau_{\text{residual, cap}}$ computed from the classical CQA output. The panel raises the Large Transient alert at operator-configured amber and red thresholds on p_{sat}, following the same two-level caution / warning pattern used elsewhere in the operator UX.

The semantics of the alert are precise. p_{sat} does not answer **what is the vessel's excursion after a WCF**; it answers **how likely is it that the linear motion prediction can be trusted**. When p_{sat} is in the green range, the output of the ensemble evaluator is a tight upper bound on the post-WCF excursion and is what the operator should act on. When p_{sat} crosses into amber or red, the ensemble output should be read as a lower bound and the operator should treat the actual excursion as materially larger than the number on the screen. The channel therefore behaves as a **trust indicator on the motion prediction**, not as a duplicate of the classical CQA capacity check.

The channel closes the "readily available measurements only" design goal of the Goal section: it consumes only the live allocator feedback and the per-DOF residual cap constant computed from the classical CQA output. It does not consume a sea state, a wave forecast, or a spectral partition. This makes it defensible under the operationally common conditions — mixed wind-sea and swell, cross-current forcing, shielding — that would break a QTF-based online estimator of the post-WCF forcing.

Two-channel operator decision surface

The ensemble evaluator and the saturation-probability channel are combined with the classical CQA output into the operator decision surface of Table 1. The table is the operator's mental model of the live panel in one view: for any combination of the three channels the operator has an unambiguous reading of what to trust and what to act on. The **Post-WCF motion** column reads the output of the ensemble evaluator relative to the intact P95: **similar** means the post-WCF prediction is close to the present intact footprint; **elevated** means it is materially larger. The **Saturation** column is the saturation-probability channel, which qualifies how far the motion number in the previous column can be trusted.

The decision surface preserves the classical alarm ordering: when the classical CQA fires Drift Off or Force Off, the classical alarm takes precedence and the motion-layer output is displayed as supporting context rather than as the operator's primary action cue. Within the classical-CQA-OK state, the two-channel readout gives the operator four distinct positions on the way toward saturation. The three operator-visible alerts of the framework are therefore:

- **Drift Off** (classical CQA warning) — unchanged from current DP-2/DP-3 practice.

Table 1: Live operator decision surface. Rows correspond to increasing operational concern.

Classical CQA	Post-WCF motion	Saturation	Interpretation
OK	similar	green	Vessel operating in a comfortable state; no action needed
OK	elevated	green	Elevated post-WCF motion, but the prediction is trustworthy
OK	elevated	amber	Motion prediction is a lower bound; treat with caution
OK	elevated	red	Motion prediction unreliable; treat as worst-case
Drift Off/Force Off	(any)	(any)	Classical alarm dominates; motion-layer output is context

- **Force Off** (classical CQA warning) — unchanged from current DP-2/DP-3 practice.
- **Large Transient** (motion CQA caution / warning) — new; fires when p_{sat} crosses its amber or red threshold, indicating that the ensemble output should be read as a lower bound.

No other alerts are added or removed; existing DP UX conventions, including audible / visual behaviour of Drift Off and Force Off, are preserved.

Long-term post-WCF motion larger than the short-term transient

The post-WCF response has two distinct phases with independent risk profiles. The **transient phase** — the first 30-60 s after the failure — is dominated by the survivor allocator absorbing the step in τ_{lost}; its peak excursion is what the classical CQA implicitly verifies against by requiring the survivor set to generate the pre-failure force and moment. The **long-term phase** is a new steady state under reduced capacity: with the surviving peak τ closer to the mean environmental forcing than in the intact condition, the LF headroom against wave-drift variability is compressed and the running-maximum footprint grows.

In some sea states the long-term P95 footprint exceeds the transient peak. The classical CQA sees the transient pass and reports green, but the operator is nonetheless outside the motion envelope. The ensemble evaluator captures this correctly: it propagates the closed-loop vessel model over a window long enough (~5 min) to cover both phases, so the reported P95 is the true post-WCF peak, transient or long-term whichever is larger. This is the regime where the motion layer is materially different from a transient-focused CQA extension, and it is one of the cases exercised in the long-term post-WCF validation.

Estimator convergence

The intact-side envelope and the transient channel both consume sliding-window estimates of process moments from live data, so the numbers on the operator screen are only as trustworthy as those estimates. Figure 8 shows the estimator settling behaviour on a stationary synthetic position-deviation channel with known true σ = 0.60 m. Three independent realisations are run through the same online estimator used on the live panel, with a prior deliberately biased high at σ = 0.90 m to make the warm-up visible. The 5-95% posterior credible band collapses onto the true value as data accumulates in the 10-minute window, and by the time the window is full the run-to- run spread is well inside the band width the operator sees on screen. The band width is the operator's cue: while it is wide the live panel numbers should be read with the extra margin the band implies; once it has tightened the numbers can be acted on as if they were the offline pre-op prediction.

The 10-minute window and the accompanying implicit assumption that the process is stationary over that window are the price of running the estimator online without a sea-state input. Both assumptions are comfortable against how sea states actually evolve in the North Sea and comparable basins: significant wave

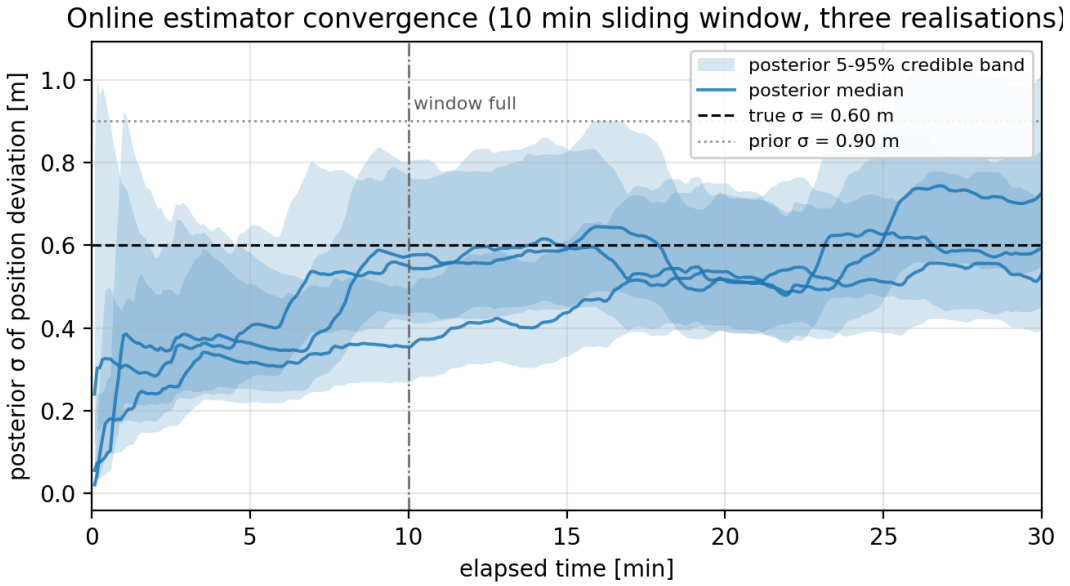


Figure 8: Online estimator convergence on a stationary synthetic position-deviation channel. Three independent realisations with true $\sigma = 0.60$ m are fed into the sliding 10-minute Bayesian estimator with a prior deliberately biased high at $\sigma = 0.90$ m to make warm-up visible. The posterior median (solid) and 5-95% credible band (shaded) converge onto the true value as the window fills; the band width itself is what the operator reads as trust in the live number.

height, peak period, and mean wind speed on a typical forecast grid change on timescales of hours, not minutes, so a rolling window an order of magnitude shorter than the forcing decorrelation time captures a locally stationary process to good approximation. Frontal passages and squall lines are the practical exceptions and are already handled outside the CQA loop by conventional weather routing.

Simulation Studies

Test vessel and simulator

The framework is validated against a benchmark CSOV ($L_{pp} = 101.1$ m, $B = 22.4$ m, $T = 6.5$ m) configured with a five-thruster layout typical of the class: two bow tunnel thrusters, a bow azimuth, and two stern main-propeller / rudder units. The reference truth is a high-fidelity nonlinear DP simulator with seakeeping-derived hydrodynamics, per-thruster force and moment allocation including allocator dynamics, and a nonlinear passive observer of the same class used in fielded DP-2/DP-3 systems. The simulator produces vessel motion at the level required to verify a live CQA panel: the position deviation $\eta(t)$, velocity $\dot{\eta}(t)$, allocator feedback $\tau_{\mathrm{fb}}(t)$, and observer states $\hat{b}(t)$, $\hat{\eta}_w(t)$ that the panel would read on a real vessel are all available for direct comparison against the panel's own predictions. The panel is exercised on the same input streams the field system would see, with the simulator supplying the reference truth.

The validation grid is the 14 cells of Table 2. The Beaufort 6 and Beaufort 8 environments are the nominal and extreme sea states tabulated in DNV-ST-0111 [10]; the half-Beaufort cells interpolate the same slope to probe the saturation boundary more finely, and the Beaufort 4 cells substitute a stronger, obliquely-set current to give a current-dominated check at low wave excitation. Two waves-only cells complete the grid, representing station-keeping driven purely by wave loading with no wind or current — the condition under

which the wave-drift term alone sets the low-frequency motion. Cell names are used directly as labels in the roll-up figures later in this section.

Two conventions in the table are worth stating. **Wave heading** is the wave direction relative to the vessel, 0° being head seas and positive values off the port bow; the vessel heading itself is held fixed across all cells. **Wind offset** is the wind direction relative to the waves, so 0° means wind, waves and current are co-linear and 45° splits the wind off the wave direction. Current runs with the waves except on the Beaufort 4 cells, where it is set 45° off the vessel heading at 1.0 m/s to load the vessel asymmetrically. Each cell is run for 30 seeds, giving 420 realisations in total.

Table 2: The 14-cell simulator validation grid.

Cell	Bf	Vw	Hs	Tp	Vc	Wave heading	Wind offset
bf4_c1_h0	4	7.0	1.5	6.00	1.0	0	co-linear
bf4_c1_q10	4	7.0	1.5	6.00	1.0	+10	co-linear
bf6_h0	6	13.8	3.1	8.50	0.75	0	co-linear
bf6_q10	6	13.8	3.1	8.50	0.75	+10	co-linear
bf6_h0_w45	6	13.8	3.1	8.50	0.75	0	+45
bf6_q10_w45	6	13.8	3.1	8.50	0.75	+10	+45
bf7p5_q10_w45	7.5	19.0	5.1	9.63	0.75	+10	+45
bf8_h0	8	20.7	5.7	10.00	0.75	0	co-linear
bf8_q10	8	20.7	5.7	10.00	0.75	+10	co-linear
bf8_h0_w45	8	20.7	5.7	10.00	0.75	0	+45
bf8_q10_w45	8	20.7	5.7	10.00	0.75	+10	+45
bf8p5_q10_w45	8.5	22.4	6.4	10.38	0.75	+10	+45
pwo	–	0.0	4.2	10.22	0.0	+90	waves only
pwq30	–	0.0	4.2	10.22	0.0	+30	waves only

Units: V_w and V_c in m/s, H_s in m, T_p in s, wave heading and wind offset in degrees.

Capability to intact motion

The left-hand panel of the motion roll-up (Figure 9) compares the framework's predicted P_{95} of the intact-condition running-maximum footprint $|R_{LF}|$ against the corresponding simulator truth on each cell. The cells span Beaufort 4 to Beaufort 8.5 across head, quartering, and beam headings, plus two waves-only variants. Across all 14 cells the framework's P_{95} is within +7% to +28% of the simulator truth, with a median error of +17%. All 14 cells are on the positive side of zero — the framework systematically overpredicts.

The systematic conservatism is not a modelling error but a known property of the method: the extreme-value result is mildly conservative for a feedback-controlled vessel, because the controller suppresses the largest excursions the underlying theory allows for. The observed bias, roughly a factor 1.15-1.20 in P_{95}, is consistent with that expectation, and its sign is the one the framework intends: the operator receives a mild upper bound rather than a mean or a lower bound.

The 14-cell agreement therefore validates both parts of the intact-side chain: the vessel-response model and the extreme-value envelope built on it. The framework's intact-side output is the same object used in the pre-op operator summary and in the live intact bar of the extreme-value section; the roll-up confirms both.

Transient at WCF

The right-hand panel of Figure 9 shows the framework's predicted post-WCF P_{95} alongside the simulator truth on the same 14 cells. On the eleven cells where the survivor allocator does not saturate (the p_sat channel remains green throughout), the framework's P_{95} is within -23% to +16% of the simulator truth, with a median error of -10%. On these cells the linear closed-loop ODE that drives the ensemble evaluator is quantitatively accurate: the transient step in τ_{lost}, the allocator absorbing that step, and the LF re-equilibration are all captured within the intrinsic scatter of the sea-state statistics.

The mild negative bias on the transient axis has a different origin than the positive bias on the intact axis. Where the intact axis sits on the built-in conservatism of the extreme-value result, the WCF axis reflects the sea-state averaging over the 5-minute post-WCF window used by the ensemble evaluator. The integrated bias is small enough — comparable to the seed-to-seed scatter within each cell — that the framework's WCF number is a tight prediction rather than an upper or lower bound. Where the prediction should be treated as a lower bound is the regime the saturation-probability channel is designed to flag, and which the roll-up covers in the long-term post-WCF validation.

Long-term post-WCF envelope

The rightmost columns of Figures 9 and 10 are the cells where the survivor allocator saturates in the post-WCF steady state — the regime the case-study of the long-term post-WCF discussion identifies as the operationally interesting one. On these cells the linear closed-loop ODE that drives the ensemble evaluator is no longer sufficient, and the framework's motion P_{95} drifts from the simulator truth. The worst case in the roll-up is bf8p5_q10_w45 (Beaufort 8.5, waves 10° off the bow, wind split 45° from the waves; Table 2): the simulator reports an LF peak P_{95} of 29.9 m, driven by seeds where the vessel loses station outright in the post-WCF long-term regime; the framework's predicted P_{95} is 5.4 m, an 82% underprediction.

The 5.4 m motion prediction is nonetheless not what the operator sees on that cell. The framework raises a red alarm on bf8p5_q10_w45 through two independent channels. First, the predicted motion P_{95} of 5.4 m already exceeds the alarm radius of 4 m used for the CSOV, so the classical CQA motion bar fires red on 28 of 30 simulator seeds. Second, the saturation-probability channel evaluates p_{sat} with a mean of 0.14 and a maximum of 0.26 on the same cell — well above the red threshold of 0.10 — and fires the Large Transient alert on 21 of 30 seeds and the amber threshold on the remaining 9. The operator therefore sees red on this cell through two independent gates before the simulator ever runs the WCF that would produce the catastrophic excursion.

This is the design intent of the two-channel layout of the operator decision surface: where the linear ODE breaks, the trust indicator triggers so that the operator is not misled by an artificially small P_{95}. The rejected-extensions section discusses three plausible refinements to the linear ODE that would in principle produce a tighter P_{95} on the saturated cells, and were investigated and rejected.

Sea-state and heading sweep summary

Figures 9 and 10 give the whole 14-cell scoreboard for the live CQA panel against the simulator truth. The cells cover Beaufort 4 through Beaufort 8.5 in head, quartering, and beam headings, plus two waves-only variants representative of purely-wave-driven station-keeping. Figure 9 covers the two motion axes the operator reads directly; Figure 10 covers the saturation channel that qualifies them.

The two operational messages of the roll-up are already stated in the intact-motion, transient, and long-term post-WCF validation sections, and are summarised here. First, on the 11 non- saturated cells the intact and post-WCF motion P_{95} are accurate to within a factor of about 1.2 in either direction, with systematic biases

(positive on the intact axis, negative on the WCF axis) that reflect the intended design conservatism and the finite- window averaging respectively. Second, on the three cells where the survivor allocator approaches or crosses saturation (bf7p5_q10_w45, bf8_q10_w45, bf8p5_q10_w45), the WCF motion P_{95} underpredicts the simulator truth by up to 82% — but the two independent gates of the operator decision surface both fire red, giving the operator the correct actionable outcome despite the imprecise motion number.

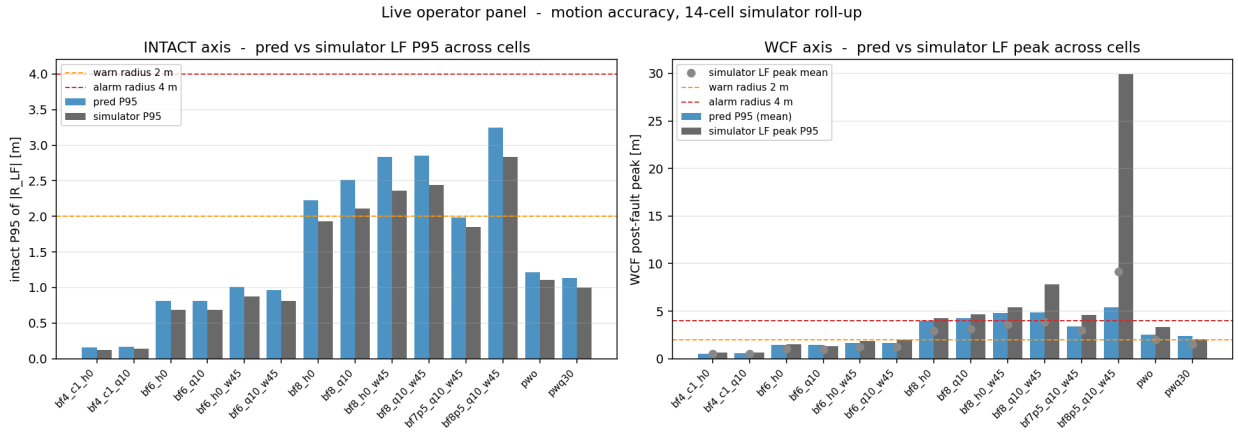


Figure 9: Motion accuracy across the 14-cell simulator roll-up. Left: intact axis P_{95} of $|R_{LF}|$, predicted (blue) vs simulator (grey), against the warn (2 m) and alarm (4 m) radii. Right: post-WCF P_{95}, predicted (blue) vs simulator LF peak (grey), with the simulator LF peak mean marked (dots).

The right-hand panel of Figure 10 shows the additional analytical saturation-excursion contribution developed as a refinement to the ensemble-evaluator output on saturated cells. On the 13 non-saturated cells the contribution is at most a few centimetres and does not materially change the operator display; on bf8p5_q10_w45 it adds 0.85 m to the reported P_{95}, keeping the red flag but not closing the underprediction. The refinement is therefore a marginal improvement in this regime, and the rejected-extensions section discusses why we did not extend it further.

Rejected extensions

Three plausible refinements to the two-channel design were investigated and set aside.

QTF-gradient estimation online. Estimating the vessel's wave-drift QTF gradient at the operational sea state, then using it inside the ensemble evaluator to tighten the post-WCF envelope. Rejected on theoretical grounds: online QTF gradient estimation requires wave-field assumptions that break in operationally common conditions — mixed wind-sea and swell, cross-current, near-installation shielding — and the framework's measurement-only design goal (see Goal) does not accommodate a runtime wave-field model.

Pre-trip variance as within-cell predictor. Using the observed variance of the pre-WCF LF motion as a predictor of the seed-to-seed scatter in the post-WCF peak, within a fixed sea state. Empirically falsified: per-cell Pearson correlations between pre-trip variance and post-WCF peak range from -0.7 to +0.7 across the 14 cells of the roll-up, with no consistent sign.

Pre-trip PSD-shape as within-cell predictor. Same idea, but using the pre-WCF Vanmarcke spectral bandwidth. Empirically falsified on the same evidence, with the same range of correlations.

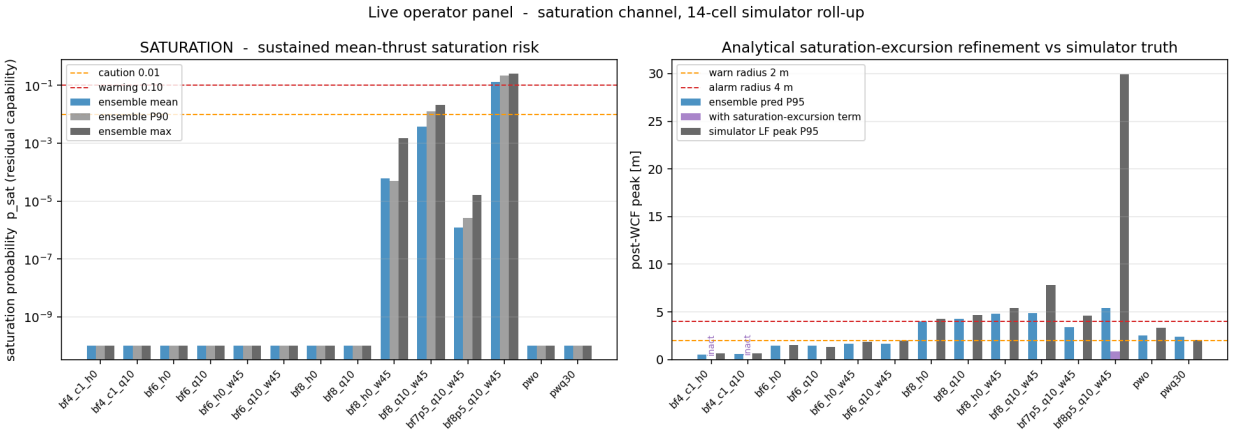


Figure 10: Saturation channel across the 14-cell simulator roll-up. Left: sustained mean-thrust saturation probability p_{sat} per cell, ensemble mean / P90 / max shown against the amber (0.01) and red (0.10) thresholds. Right: the analytical saturation-excursion contribution on top of the ensemble-evaluator output, alongside simulator truth for reference.

The two-channel design is therefore the **tight** envelope on this class of vessel and instrumentation: adding observables from past history to characterise the future wave-phase-at-trip alignment does not add predictive power.

Conclusion

This paper has developed a motion-based consequence quality analysis (CQA) framework that closes the motion gap in the classical DP capability and CQA machinery. The contribution is two-part. The pre-operation risk assessment (Motion-Based Capability Analysis) extends the classical capability analysis with an estimate of vessel and mission-equipment motion from the forecast sea state, delivered as the operability polar, the WCFDI decision matrix, and the pre-op operator summary. The live CQA panel (Consequence Analysis (Live)) reformulates the same underlying model for online execution, delivering intact and post-WCF motion P_{95} continuously from real-time measurements and the DP model, with a saturation-probability trust indicator that fires when the linear motion prediction is no longer reliable.

The empirical basis is a 14-cell simulator validation across Beaufort 4 to Beaufort 8.5 (Simulation Studies): the intact-side P_{95} is accurate to within +7% to +28% (systematically conservative by design), the post-WCF P_{95} is within -23% to +16% on the eleven non-saturated cells, and the two-channel operator decision surface correctly red-flags the three saturated cells through two independent gates even where the motion prediction itself is understated. One limitation of this evidence should be stated plainly: the simulator generates its sea states from the same spectral description the prediction model consumes, so the intact-side agreement tests the motion and statistical machinery but not the spectral assumptions themselves. Confirming those against measured vessel motion is the outstanding validation step. Validation against logged operations from in-service vessels is left to future work.

The framework preserves the operator's mental model — motion, footprint, proximity to limits — rather than translating the DP system's actions into force reserves that the operator then has to map back to expected motion. The classical Drift Off and Force Off alarms remain the primary hard-failure signals; the framework

adds one motion-based alert (Large Transient) that tells the operator whether the motion forecast can be trusted as a tight upper bound or should be read as a lower bound.

The framework is deliberately additive to the classical rules. The capability + CQA computation mandated by IMO MSC/Circ.645 [1] and the class rules [2, 3], and the station-keeping limits established for the operation under guidance such as DNV-ST-0111 [10] and IMCA M254 [4], are not superseded or contested; applying a percentage capacity margin on top of the classical CQA continues to land the operator comfortably inside a safe operating envelope for the majority of operations. What the motion layer adds is the ability to work closer to the operational limit in a controlled way — the operator sees continuous, graded, motion-based proximity to the limit rather than waiting for a breach. This directly enables the walk-to-work and heave-compensated operational regimes that have expanded the DP envelope beyond simple hold-vs-drift.

The measurement-only online scope set out in the Goal section is honoured throughout: the live layer consumes only signals every DP-2/DP-3 vessel already publishes, plus vessel-constant parameters. No online seakeeping model, no runtime spectral partition, no runtime wave-field assumption. This is a deliberate scope limit that keeps the design defensible under operationally common mixed sea states, cross-current forcing, and near-installation shielding effects that would break wave-field-based online refinements (see Rejected extensions).

Future work has three lines. First, validation of the intact side against logged operations from in-service DP vessels, comparing the forecast-driven motion envelope against realised vessel motion with the archived forecast cross-checked against hindcast reanalysis. This is the step that tests the spectral assumptions outside the simulator that shares them, and it is the natural route to the field evidence needed to support rules-alignment discussions. Second, extension of the mission-equipment coverage to further motion-sensitive systems (LARS, heave-compensated cranes, ROV winch handling). Third, presentation of the motion-based CQA deliverables to the rulemaking bodies as candidates for inclusion in the next revision of IMCA M254 and the DNV-ST-0111 transient analysis guidance — a natural next step given that the current guidance already gestures in this direction (see Introduction) but stops short of requiring the delivery.

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(TODO: whoever helped)

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